

Use `tf.convert_to_tensor` for `symbol_values` in differentiator docstrings

<https://github.com/tensorflow/quantum/issues/427>

ROW 130

Use tf.convert_to_tensor for symbol_values in differentiator docstrings #427

[New issue](#)[Closed](#)

#494



therooler opened on Nov 5, 2020

...

Python: 3.8
TFQ: 0.4.0

I was trying to see what kind of speed up I would get from using `tfq.differentiators.Adjoint`. The [docstring example](#) runs fine, but when I add more parameters to the circuit, I get the following error message

```
Traceback (most recent call last):
  File "/home/rooler/Documents/tfq/kagome/does_adjoint_work.py", line 27, in <module>
    grads = g.gradient(expectations, symbol_values_tensor)
  File "/home/rooler/anaconda3/envs/tfq/lib/python3.8/site-packages/tensorflow/python/eager/backprop.py", line
    flat_grad = imperative_grad.imperative_grad(
  File "/home/rooler/anaconda3/envs/tfq/lib/python3.8/site-packages/tensorflow/python/eager/imperative_grad.py",
    return pywrap_tfe.TFE_Py_TapeGradient(
  File "/home/rooler/anaconda3/envs/tfq/lib/python3.8/site-packages/tensorflow/python/ops/custom_gradient.py",
    raise ValueError(
ValueError: ('custom_gradient function expected to return', 5, 'gradients but returned', 4, 'instead.')
```

Code to reproduce:

```
import cirq
import tensorflow_quantum as tfq
import tensorflow as tf
import sympy
import numpy as np

my_op = tfq.get_expectation_op()
adjoint_differentiator = tfq.differentiators.Adjoint()
# Get an expectation op, with this differentiator attached.
op = adjoint_differentiator.generate_differentiable_op(
    analytic_op=my_op
)
qubit = cirq.GridQubit(0, 0)
circ = cirq.Circuit(cirq.X(qubit) ** sympy.Symbol('alpha'))
circ.append(cirq.Y(qubit) ** sympy.Symbol('beta'))
circuit = tfq.convert_to_tensor([circ])

psums = tfq.convert_to_tensor([[cirq.Z(qubit)]])
symbol_values_array = np.array([[0.123, 0.3]], dtype=np.float32)

# Calculate tfq gradient.
symbol_values_tensor = tf.convert_to_tensor(symbol_values_array)
with tf.GradientTape() as g:
    g.watch(symbol_values_tensor)
    expectations = op(circuit, ['alpha', 'beta'], symbol_values_tensor, psums
)
grads = g.gradient(expectations, symbol_values_tensor)
print(grads)
```

I am having some trouble trying to debug this. Any suggestions?



zaqqwerty on Nov 5, 2020 · edited by zaqqwerty

[Edits](#) [Contributor](#) ...

I reproduced this error with this code in colab. I find that if I replace `['alpha', 'beta']` with `tf.constant(['alpha', 'beta'])`, the error goes away. As you say, I also find the original code snippet in the Adjoint Diff docstring still works. We should get to the bottom of this, but for now making sure all op inputs are tensors seems to fix it



Assignees

MichaelBroughton

Labels

No labels

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

[Fix diff docstrings.](#)
tensorflow/quantum

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MichaelBroughton on Nov 5, 2020 · edited by MichaelBroughton

Edits

Collaborator



This was actually an undocumented backwards breaking change in regular TensorFlow (I spoke with an engineer about this when we first upgraded from 2.1.0 to 2.3.1). Normally it would be fine to pass a python list of strings in, but since the change we should recommend/encourage users to pass in constant tensors. You can see in here when we did the upgrade that we also needed this change internally as well: <https://github.com/tensorflow/quantum/pull/340/files>.

Let's convert this issue into: Differentiator docstrings do not convert to tensor for symbol_values and then we can PR a fix for everyone.



MichaelBroughton changed the title **Adjoint differentiator does not return enough gradients. Use `tf.convert_to_tensor` for symbol_values in differentiator docstrings** on Nov 5, 2020



therooler on Nov 5, 2020

Contributor

Author



Changing to `tf.constant` did the trick. Thanks.



MichaelBroughton self-assigned this on Feb 25, 2021



MichaelBroughton mentioned this on Feb 25, 2021

Fix diff docstrings. #494



MichaelBroughton closed this as **completed** in **#494** on Feb 25, 2021



jaeyoo added a commit that references this issue on Mar 30, 2023

Merge pull request [tensorflow#427](#) from cognigami/docs-refresh



Verified

dbd6a68

Fix diff docstrings. #494

<> Code

Merged MichaelBroughton merged 4 commits into master from adjoint_doc on Feb 25, 2021

Conversation 4 Commits 4 Checks 0 Files changed 3 +18 -15

**MichaelBroughton** commented on Feb 25, 2021 Collaborator

Fixes #427

Fix diff docstrings. 7e1634c

MichaelBroughton requested review from zaqqwerty and jaeyoo 4 years ago

Merge branch 'master' into adjoint_doc Verified 82af235

**zaqqwerty** suggested changes on Feb 25, 2021 View reviewed changes

zaqqwerty left a comment Contributor

small change

tensorflow_quantum/python/differentiators/adjoint.py Outdated Show resolved

MichaelBroughton added 2 commits 4 years ago

A feedback. df4ed60

Merge branch 'adjoint_doc' of github.com:tensorflow/quantum into adjoint_doc 41583b2

MichaelBroughton requested a review from zaqqwerty 4 years ago

**zaqqwerty** approved these changes on Feb 25, 2021 View reviewed changes

zaqqwerty left a comment Contributor

LGTM

Reviewers

 zaqqwerty ✓

 jaeyoo ●

Assignees

No one assigned

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Development

Successfully merging this pull request may close these issues.

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